# Ibasho agent blackboard

## State (orchestrator, kept current)

**orchestrator** state · active
**Phase:** M-A first light plus independent M-B work, light testing. Full history: `docs/archive/BLACKBOARD-to-M0.typ`.
**Done:** G0 scaffold, M-0 slice (physics, rules, search and money, Lean spec and checker with TS mirror, narrator and memory, scene, agent link, first app screen, content and landing, hero assets), SPEC 14 contracts in core.
**In progress:** WP-01 physics (A1 to A10, maintenance decay), WP-06 UI (M-A and M-B components), WP-07 scene (review), WP-08 app (shell, onboarding, Reveal with scene), WP-03 rules (M-B), WP-02 money and score (M-B, SPEC 14 run and negatives), WP-04 checker gaps (bills, score, memory.completed).
**Idle until M-D:** WP-05 AI, WP-09 content, WP-12 agent link, WP-10 QA (testing is light for now), WP-11 assets.

## Contract index (where the source of truth lives)

- Contracts and zod schemas: `@ibasho/core` (`packages/core/src/**`, `@ibasho/core/schemas`). SPEC 14 kinds: `maintenance`, "If nothing is done" run, per-package `negatives`.
- Measure and option ids: `packs/home-uk/src/ids.ts` (frozen; GH1 432 heavy, 10,800 packages).
- Entry points: packs export `./engine`, `./ui`, `./ids`; apps start at `src/main.tsx`; `packages/fixtures` exports `./*` (no index).
- Certificate, constants, seals, spec hash: generated from Lean into `@ibasho/spec-gen` (real, not a placeholder).
- Ownership as code: `tools/lint/ownership.ts`. `ACKNOWLEDGEMENTS.md`, the blackboard, STATUS and `docs/refs.yml` are shared append.
- Agent link: `node apps/mcp/src/main.ts` (no build); `pnpm agent:install`.

## Decisions (condensed; details in the archive)

**orchestrator** decision · final

- Stack: pnpm monorepo, Vite, React, TypeScript 6.0.3, R3F, IndexedDB, Biome, Lean 4 v4.34.0 core only. Local first, offline first. Versions in ARCH 13.
- Plan of record: VISION, SPEC (rulings R1 to R24), ARCHITECTURE, DESIGN_SYSTEM (Andon), BUILD_PLAN; red-team rulings in `docs/review/DECISIONS.md` override older text.
- Hero home GH1: 1905 London mid-terrace in a conservation area. Score weights F 1/5, C 1/5, E 3/20, A 3/20, H 1/10, R 1/10, T 1/10, from Standard mode; shown as a number with four groups, no letter. Futures: Least upfront, Best return, Lowest running carbon.
- Lean owns formulas and predicates; pack data travels in certificates; the checker is proved sound and the TS mirror matches it.
- Engine determinism: no inexact `Math` in engine code, `dmath` instead, fixed point after quantise.
- Narrator is a plugin: template always on, Ollama local (`qwen3:8b` default; `:cloud` models count as cloud and go in the outbound ledger), Claude later.
- Agent link (WP-12): MCP server, bridge mounted by the companion, Claude Code plugin, installer. Companion: loopback host, Origin allow list, Bearer token for scripts.
- Bad outcomes (SPEC 14): "If nothing is done" line, custom what-if requests as typed ops, negatives beside savings.
- Reuse proven open source and credit it in `ACKNOWLEDGEMENTS.md`. Asset licensing is not a constraint.
- Testing is light for now: each WP runs only its own package tests. No gates until the owner asks.

## Claims

**steward:WP-00** claim · done
SPEC 14 contracts in `packages/core/**`: the `maintenance` plugin kind, the "If nothing is done" run, per-package negatives, and the deferred WP-02 `OwnerCtx` fields.

**builder:WP-08** claim · active
Gate M-A (app shell and store): `apps/web/**` except `apps/web/public/assets/**`, `packages/store/**`. Shell layout zones (left rail, top bar with home switcher and lantern, mobile tab bar), `/app/start` onboarding in 5 typed

steps with no budget question, Reveal on the home stage with WP-07 `Stage` (SVG fallback without WebGL), packs imported by package name.

**builder:WP-06** claim · active
Gates M-A and M-B UI kit: `packages/ui/**`. M-A: `Sheet`, `NarratorStrip`, `ScoreGroups`, `EpcChip`. M-B: `Futures` (with the fourth line "If nothing is done" and a negatives list on every card), `MeasureCard`, `FuelBars`, `Slider`, `WhyNot`, `CheckLane`, `MonthRing`, plus the chart frame with `TableAlt`, a first mon set and `GradeBadge`. Stories for each.

**builder:WP-02** claim · active
M-B pieces without new physics: score with exact Q and the four groups, EPC band proxy, IRR, paybacks, loan stub, flips, the "If nothing is done" run and per-package negatives (SPEC 14), costs for every SPEC 9 option. Paths: `packages/engine/src/{api,index}.ts`, `packages/engine/src/{search,finance,score,flips,evidence}/**`, `packages/engine/{test,scripts}/**`, `packs/money-uk/**`, `packages/fixtures/src/results/**`.

**builder:WP-03** claim · done
M-B (C1 to C7): `packages/engine/src/rules/**`, `packs/rules-uk/**`, `packs/rules-in/**`. Named functions, `noise()`, the SPEC 9 `rules-uk` set with `APPL.*`, sentences, obligations, rule lint.

## Interfaces (live only)

**builder:WP-02** interface · open
**Requests (non-blocking):** (1) WP-03: `BUILTIN.MEMORY` lives in the search (`packages/engine/src/search/run.ts`) because the evaluator has no memory facts; move it into `BUILTIN_RULES` when a fact form exists, keeping the id. The budget rule receives the gross central cost as `p.capex`; say if Lean `Budget.lean` should compare N0 instead. (2) WP-01: heat pump costs size the unit with a cost-side estimate (`heatPumpSizeKw` in money-uk); expose `hp.size_kw` to cost plugins at M-A, e.g. a `size` argument or a twin field. (3) WP-00: `OwnerCtx` needs fuel escalation scenarios and a confirmed counterfactual replacement (A_cf) by M-B; M-0 uses 0% escalation and no A_cf.

**builder:WP-09** interface · open
**Chapter `requires` convention:** a requirement is a plugin id or `kind:<PluginKind>` (any plugin of that kind, checked with `registry.all(kind)`); `presenceOf(registry)` adapts a `PluginRegistry`. Current landing requires: drawn `climate-london` + `kind:archetype`; what-if `loft`, `ashp`, `kind:tariff`; futures `kind:search-strategy`, `rules-uk`; proof `kind:checker-rung`. **Request to WP-08:** the landing discovers plugins at build time from `packs/*/src/{engine,ui}/module.ts` (`apps/landing/src/discover.ts`), because `apps/web/src/manifest.ts` is still empty. Once the manifest lists the registered modules, please keep it importable from Node (plain `.ts`, no JSX) or export the module list as data, and WP-09 will switch to it so the page never advertises a pack the app does not load. **Links:** "Step into your home" goes to `/app/start`; "See a sample home" goes to `/app/home/gh1` (GH1 read-only, Story mode). WP-08: tell me if the sample route differs.

**builder:WP-09** interface · open
**To WP-10 (smoke):** `apps/landing` now prerenders the real page: `tests/smoke/boot.spec.ts` will fail on `main[data-andon="blank"]` being empty. Please assert `main[data-andon="landing"]` visible, `section[data-chapter="hero"]` present, title `Ibasho`, body background still `rgb(244, 238, 227)` in light scheme, no console errors. **To WP-06 (fonts):** `@ibasho/ui/fonts.css` compiles to 848 `@font-face` rules (714 kB CSS); the landing loads it after first paint to keep AC-L1, but a Latin-first subset entry (for example `./fonts-latin.css`) would help both apps. Small woff slices also get inlined as data URIs by Vite's default `assetsInlineLimit`; the landing turns that off for fonts.

**builder:WP-12** interface · open
**Request to WP-05 (companion mount, one line plus discovery):** add `@ibasho/agent-bridge` to `apps/companion` dependencies at the next install window, then `app.route(BRIDGE_MOUNT, createBridgeApp({ hub, token }))` with `token = createServerToken()` and `hub = new AgentHub({ onLedger })`; bind `BRIDGE_HOST` (127.0.0.1); on listen `writeServerJson({ port, token, pid: process.pid, startedAt, bridge: BRIDGE_MOUNT })` and print `hub.pairingCode().code`; `removeServerJson()` on clean exit. All exported from `@ibasho/agent-bridge/server`

(constants from `@ibasho/agent-bridge`); snippet in `packages/agent-bridge/README.md`. The SSE transport needs no extra dependency; `/link/ws` turns on if you pass `upgradeWebSocket` from `@hono/node-ws`.

**builder:WP-12** interface · open

**Request to WP-08 (Settings > Agent link):** add `@ibasho/agent-bridge` to `apps/web` at the next install window. Toggle off by default with one line saying a connected agent may send what it reads to its model provider (SPEC 13); a code field; then `connectAgentLink({ origin: 'http://127.0.0.1:7357', pair: { code }, client: { app: 'ibasho-web', version }, handlers, onStatus, onLedger })`. `handlers` implements `LiveHandlers` (`packages/agent-bridge/src/protocol.ts`): `apply` runs ops through the OpValidator and the undo stack with an "agent" mark; `say` runs the narrator validator; `screenshot` returns a base64 PNG of the stage; `thread` draws the red thread. Call `link.pushState(state)` on route, chapter, layer and selection changes; keep `link.session` in sessionStorage to reconnect after a reload (`pair: { session }`); write every `onLedger` entry to the outbound ledger (SPEC 5.14). Every incoming call is already schema-checked in the client.

**builder:WP-04** interface · proposed

**To WP-02 (engine emitter), from WP-04:** the certificate now has optional slots the checkers need (`@ibasho/spec-gen cert.ts`): `groups[].optionMeta` (`{ [optionId]: { measure, kind } }`), `context.home` (`{ region, facts: homeFacts(twin), checks: twin.checks }`), `claimed.saving` (int64 [row][fuel] annual Wh, before minus after) and `claimed.carbonAnnual` (int64 [row] g, `carbonSavingG`). Until the engine emits them, `@ibasho/verify emit(result, { optionMeta, home })` fills them (saving and carbon from the certificate physics). Please emit them from `emitCertificate` so the checks test the engine's own claims. Also: `SPEC_HASH` is now real (`ab8d651f...`), so the result hash changed; run `pnpm --filter @ibasho/engine gen:results` to refresh `packages/fixtures/src/results/gh1-m0.search.json` (engine `results.test.ts` and mcp `l2-engine.test.ts` are red until then).

**reviewer:WP-04** interface · open

**To WP-02 and WP-04 (M-P): `context.memory.completed`.** SPEC AC-P1 says a measure marked done leaves future searches. Today the engine copies `memory.completed` into the certificate but does not read it, and neither `ibasho-check` nor the mirror checks it, so a certificate offering a completed measure passes. Proposal: once WP-02 defines the semantics (exclude like `rejected`, or fold into the twin), WP-04 adds it to `deriveRow` in Lean and the mirror, with a mutant.

**builder:WP-04** interface · proposed

**To WP-02 (engine emitter), checker gaps closed:** the checkers now test three more claims. (1) `money.bills`: `claimed.bills`, int64 [row][fuel] annual your-home bill in micro-pence (slab-exact, `billOf` of the row's delivered energy, standing charge rule as in your certificate entry). Missing means the check fails, so `@ibasho/verify emit` fills it from the certificate physics until `emitCertificate` does. (2) `score.recompute`: optional `claimed.score = { zone: 'uk' | 'in', mode: 'burden' | 'per-area', rows: [{ row, indicators, sub, groups, total }] }`, rationals as `Q.toWire` strings, rows strictly increasing, and every front and pick row present; sub, groups and total must equal `subScore`, `groupScore`, `score` of the indicators exactly. Without it the check is skipped with its reason. `emit(result, { ..., score: { zone, mode, indicators } })` and `scoreClaim(...)` build the block from indicators until the engine emits its own. (3) `context.memory.completed` (SPEC AC-P1): an option id, or an option of a measure id, in `completed` is excluded exactly like `rejected` (status blocked, `BUILTIN.MEMORY`). The engine copies `completed` but does not exclude it yet, so a real search with a done measure now fails `feas.recompute`; please treat `completed` like `rejected` in search. This answers the reviewer entry on `memory.completed`.

**steward:WP-00** interface · final

**Bad outcomes contracts (SPEC 14), additive, in `@ibasho/core` (`packages/core/src/maintenance.ts`):**

- **`maintenance` plugin kind** (engine, numeric, so provenance with sources and grade is required). `MaintenancePlugin`: `ages` (`{ kind: 'measure', measureId }` or `{ kind: 'system', category, systemIds? }`), `neglect` (message key), `effects: DecayEffect[]`, optional `endOfLife: { ageYears: { lo, mid, hi }, replaceWithSystemId?, explain }`. `DecayEffect`: `param` (JSON Pointer into the twin, as in `TwinPatch`), `unit`, `rate` (`{ form: 'relative', ppbPerYear: { lo, mid, hi } }` with bigint ppb, same recurrence as discounting, negative means the value falls; or `{ form: 'absolute', perYear: { lo, mid, hi } }` in `Q`), `onsetYears?`, `onsetFrom?: 'install' | 'as-of'`, `clamp?: { min?: Q, max?: Q }`, `quantity` (the evidence quantity id of the

aged value), explain. Registry check maintenance (checkMaintenance): ordered ranges, pointer params, a target, effects or an end of life. WP-01: decay models go in packs/home-uk/src/engine/maintenance/.

- **"If nothing is done" run:** optional EngineApi.doNothing?(req: DoNothingRequest): Promise<DoNothingResult>. Request: twin, owner, asOf, mode? (default your-home), horizonYears? (default owner.horizonYears), reportYears?, rate?: 'lo' | 'mid' | 'hi'. Result: hash, specHash, maintenanceFingerprint, mode, horizonYears, years: { year, deliveredWh: per fuel, bill, carbon, score? }[], effects: { maintenanceId, param, year, evidence, band? }[], replacements: { maintenanceId, dueYear: { lo, mid, hi }, cost?, evidence }[], totals: { costNpv, carbon }, negatives, notApplicable (model ids that matched nothing), graph. Every value is an evidence id, never a raw number (AC-B1). WP-02 implements, WP-08 draws the fourth line, WP-12 may expose it.
- **Negatives:** Negative { indicator, kind: 'fuel-rise' | 'money' | 'carbon' | 'performance' | 'disruption' | 'risk', evidence, before?, after?, risk?: 'moisture' | 'overheating' | 'noise' | 'fire' | 'structural' | 'other', ruleId?, maintenanceId?, message }; NegativesByPackage = Record<packageId, Negative[]>. SearchResult.negatives? covers baseline, front and picks (like graph); optional EngineApi.negatives?(resultHash, packageIds) for the rest. AC-B3 reads these. WP-09 writes the message copy.
- **WP-02 request (3), now applied:** OwnerCtx.escalation?: { scenario: 'low' | 'central' | 'high', ppbPerYear?: per fuel } (EscalationScenario, ESCALATION_SCENARIOS; scenario values belong to the money pack) and OwnerCtx.counterfactual?: { replacementDue: Tri, systemId?, cost? } (A_cf counts only on yes, DEC SC-F5). Absent fields keep M-0 behaviour. SystemCategory is now a named type.
- New EngineApi members are optional so current implementers still compile; they become required at M-B. PLUGIN_KINDS has 32 entries.

**builder:WP-07** interface · final

**Scene entry for the app shell (WP-08).** import { HouseStage } from '@ibasho/scene'; <HouseStage twin={twin} quantities={q} style="washi" /> fills its parent (give it a sized box). style: washi, yoru or svg (no WebGL). Optional: result (quantities then come from result.graph), layers (default heat-flow, zone-tint), pencil (underdrawing while computing), reducedMotion, tierCap, llmGenerating, store (shared selection), showOutline, assetsBase (default /app/assets/, null stays procedural) and stage (any other Stage prop). This replaces apps/web/src/home/HouseDrawing.tsx at M-A. **Pack ui (@ibasho/pack-home-uk/ui):** homeUkUiModule (default export of src/ui/module.ts, id pack-home-uk/ui) loads 16 measure-ui plugins, ids home-uk.measure-ui.<measureId>; mon values are keys of MONS in packages/ui/src/mon/set.ts (WP-06: please export MONS or isMonId from @ibasho/ui so a contract test can check them). FoldSpec.targets are element kinds plus system for plant drawn as props. ArchetypeThumb and thumbnailDrawing(form) draw every BuiltForm as an ink elevation (WP-08 Shape step can use them instead of local thumbs). WP-10: the ARCH 4.3 contract "every measure id has a measure-ui" can import HOME_UK_MEASURE_UI.

**builder:WP-02** interface · final

**M-B engine outputs (WP-08, WP-09, WP-10, WP-12):** EngineApi.flips(hash, param) returns every crossing (at is p/kWh to 0.001 or percent to 0.001; a pick that empties reads none); doNothing(req) and negatives(hash, ids) are implemented. SearchResult.negatives covers the baseline, front and picks. New evidence quantities (your-home unless noted): acct.saving.<fuel> (annual), money.bill_saving.<fuel>, fin.payback_simple (exact years or cat immediate/never), fin.payback_discounted, fin.irr (exact percent or cat n/a:<reason>), fin.loan_monthly, fin.cash_positive_m1, risk.<kind>.<ruleId>; standard: score.sub.<F..T>, score.group.<g>, score.total (exact, or an interval while T is unknown), score.delta; rating: epc.sap_est, epc.band_est (cat C (B to D)). Do-nothing quantities: <effect.quantity>.<modelId>.y<t> (with .lo/.hi), donothing.delivered.<fuel>.y<t>, donothing.bill.y<t>, donothing.carbon.y<t>, donothing.replacement_due.<id>, donothing.replacement_cost.<id>, donothing.cost_npv, donothing.carbon_total. Negative message keys for WP-09: negative.fuel-rise.<fuel>, negative.bill-rise.<fuel>, negative.year1-cost, negative.npv, negative.no-payback, negative.carbon-rise, negative.lifetime-carbon-rise, negative.score-fall, negative.band-fall, negative.risk.<kind>, negative.donothing.{fuel-rise.<fuel>,bill-rise,carbon-rise,replacement-due}, plus each decay effect's own explain key. The epc-band-at-least target now works (central band).

**builder:WP-02** interface · open

**Requests from WP-02 (non-blocking):** (1) WP-04: spec-gen writes integer rationals as "4/1", which core Q.fromWire rejects; the engine parses them with specQ. Either emit "4" or say this is intended. Also export prices/reference-v1 (SAP 10.2 Table 12 prices, standing charges, primary energy factors) so packages/engine/src/score/reference.ts can go. The certificate does not yet carry A_cf or escalation; both are absent unless

the owner sets them, so M-0 certificates are unchanged. (2) WP-00: `EpcProxy.sap(run)` needs the floor area (money-uk adds `sapFor(run, tfaM2)`); a money-pack kind for escalation scenarios (values now SF 5.1 defaults in `finance/owner.ts`); a disruption field on `MeasureModel` (days, rooms) for disruption negatives. (3) WP-01: put PV yield and self use on `SimResult` (the R indicator uses exported energy as a lower bound); write the decay models in `packs/home-uk/src/engine/maintenance/` (the engine runs any `maintenance` plugin now; tests use a boiler model of −1% a year, end of life 12, 15, 20 years).

**builder:WP-03** interface · final

**Rules M-B for WP-02, WP-04, WP-08, WP-12** (all from `@ibasho/engine`, source `packages/engine/src/rules`). `evaluateRules` now uses `DEFAULT_FNS` (pass `fns: {}` for the old all-unknown behaviour), merges `K_FACTS` (`k.*`) into every env and `derivePackageFacts(options, home)` under `ctx.pkg` for package rules (`p.kind.<kind>`, `p.adf.{major,minor,now}`, `p.vent.provided`, `p.airtighter`, `p.export.{W,limitW}`, `p.envelope.share`, `p.funded.pas2035`); `packageEnv(ctx)` returns the exact package env. `homeFacts` covers RP 4 (`h.wall.u`, `h.wall.solidMasonry`, `h.art4.P14.A` style codes, `h.dhw.class`, `h.vent.ap50`, `h.floor.type`, `h.envelope.areaM2`, `h.past.adf.*`, …). `packageFacts` takes `fundedPas2035?`. Built-ins add `BUILTIN.TENURE.{TENANT,LEASE}`, `BUILTIN.PREREQ.BATTERY-PV`, `BUILTIN.CONFLICT.{EWI-IWI,GLAZING}`. Also: `normaliseObligations`, `obligationPatch(twin, obligations)` (`{ patch, unpatched }`, idempotent), `renderRule`/`renderExpr`, `lintRules`, `noise({ twin, units })`, `noiseFact(twin, units)`, `mcs020aLpLe`, `adfCategory`, `meesCapMp`, `dtHosting`, `ensRetv`.

- **WP-02:** (1) wire `EngineApi.noise` in `src/api.ts` to `noise` from `./rules/index.ts` (it returns `NoiseResult` synchronously). (2) Apply obligations with `obligationPatch` before re-running physics; map `{ kind: 'assessment', check: 'moisture-assessment' | 'pas2035-assessment' }` to the costed options `assess-moisture` and `assess-pas2035`; `emitter-flow` and `hp-meets-design-load` come back in `unpatched`. (3) Pass `fundedPas2035: true` to `packageFacts` only when a stacked grant requires PAS 2035. (4) The rule pack grew, so regenerate `gh1-m0.search.json` (statuses are unchanged; outcomes and consents grow). (5) `BUILTIN.MEMORY` and `BUILTIN.TARGET` stay in the search.
- **WP-04 and WP-02 (certificate parity):** package rules now read derived `p.*` facts, and `APPL.*`/built-in blocks depend on them. The certificate should carry `packageEnv(ctx)` (or the derived facts) per row so `feas.recompute` sees the same env. Named functions are used only in `consent`/`oblige`/`warn`/`info` rules (lint `fn-status`), so status never depends on `NAMED_FN_IMPLEMENTED`. Lean mirrors to add when convenient: `mees_cap`, `dt_hosting`, `ens_retv` (TS is exact rationals; contracts in `fns/*.ts` headers).
- **WP-01:** emit the option facts listed in `OPTION_FACTS` (`@ibasho/pack-rules-uk/engine`), above all `m.ashp.noise = noiseFact(twin, units)`, `m.ashp.{volumeM3,units,mount,frontsHighway,nearerHighway,coolingOnly}`, `m.wall.share`, `m.ewi.{finish,matches}`, `m.pv.{mount,exportW,frontsHighway,projectionM}`, `m.battery.exportW`. Until then those rules read unknown and list their consent (worst case), which is honest.
- **WP-09:** message keys used by rules are `rules.<rule id lowercased>.explain` plus each effect's `reason`/`risk`/`note` key; `renderRule` gives a fallback English sentence.

## Blockers

**builder:WP-12** blocker · open

Update to the WP-12 blocker: L2 is no longer blocked (green against the real engine). Still waiting: L5 live e2e (WP-05 mount, WP-08 Settings and handlers), `ibasho_verify` mirror (WP-04), L8 evals need a run of Claude Code against the plugin.

**builder:WP-02** blocker · open

WP-01 physics edits in progress break two engine tests: `light path differs from full simulation for heat-ashp+loft-300 in rating` (the heavy and light split no longer matches a full run) and fixture drift. With committed physics the engine suite passes (136 tests). WP-01: when the physics lands, check the factorisation test and run `pnpm --filter @ibasho/engine gen:results` (or ask WP-02).

## Handoffs

**steward:WP-00** handoff · done

**SPEC 14 contracts landed** in `@ibasho/core`: `maintenance` kind and `MaintenancePlugin`, `DoNothingRequest`/`DoNothingResult`, `Negative`/`NegativesByPackage`, `SearchResult.negatives?`, optional `EngineApi.doNothing?` and `negatives?`, `OwnerCtx.escalation?` and `counterfactual?`. See the Interfaces entry. Core tests 140 pass;

typecheck of core and its 22 dependants passes. Not done here: decay models (WP-01), the run itself (WP-02), the fourth line (WP-08), no zod schema for `DoNothingRequest` yet (add when a wire boundary such as MCP needs it).

**builder:WP-07** handoff · done

**Scene, M-A slice.** What exists: the stage (Washi and Yoru still life, SVG fallback and pencil), procedural house from the twin, heat-flow motes, thermal lens, zone tints, roof lift, picking, `a11y-scene` outline, quality ladder, and now `HouseStage` (see Interfaces) and the `home-uk` ui half (`measure-ui` for all 16 measure ids, archetype thumbnails for all 9 built forms). Look fixes: gables are plaster on the inside, so the far wall of the cutaway reads as a room; the stage paper grain is lighter (16 fibres, lower opacity). Tests: `@ibasho/scene` 10 files, 59 passed (new: cutaway composition, house build invariants, `HouseStage` defaults); `@ibasho/pack-home-uk` 4 files, 30 passed (new `test/ui`, 7). Typecheck green on both; Biome, dependency-cruiser and `tools/lint` clean on owned paths. Not built yet: M-B layers (ghost, noise rings, fold assembly, split, miniatures), people, `packs/home-in/src/ui` (T3). G7 frame rate on Iris Xe still to be recorded.

**builder:WP-08** handoff · done

**M-A shell, onboarding, Reveal.** `/app/start` runs five typed steps (`apps/web/src/start/steps.ts`: time budget guard, refuses any money question, R24). Offline place list and local postcode area read. Every unanswered value is an estimate and marked (AC-G2). `/app/` goes to the last own home, else `/start`. Reveal navigates to `/app/home/:id` with `state.reveal`; `apps/web/src/home/HomeHouse.tsx` mounts `@ibasho/scene` `HouseStage` (washi or yoru from the lantern, pencil while computing, `ROOF_LIFT` once when ready, skipped under reduced motion); no WebGL or a stage error falls back to the SVG house, pencil then ink. Layout zones are one component, `apps/web/src/shell/Zones.tsx` (Niwa, Engawa, Zashiki, Tokonoma). `manifest.ui` now loads `@ibasho/pack-home-uk/ui`; all packs by package name. Stubs: `start/seed.ts` seeds from the GH1 survey twin with every value re-tagged as an estimate until WP-01 `ArchetypePlugin.seed()` exists (only the terrace and Not sure tiles are pickable); all places use London weather; Tune order is fixed until `valueOfInformation`. Tests: `apps/web` 14 passed (3 files, incl. a seeded home reaching the ready state on the real engine worker body), typecheck and Biome clean. For WP-10: DOM hooks `.ib-home[data-state]`, `.ib-house[data-renderer]`, `.ib-choices[data-estimated]`.

**builder:WP-06** handoff · done

**UI kit, M-A and M-B sets.** `import '@ibasho/ui/andon.css'` (now also pulls `kit.css`). M-A: `Sheet` (side sheet from 900 px, bottom sheet with peek, half, full; every snap has a button; Esc closes; not modal), `NarratorStrip` (sentences settle one by one; "Plain narrator" marker; chips without seal glyphs, R11; `sentencesFromRendered(RenderedNarration)` adapts WP-05 output), `ScoreGroups` (numeral 0 to 100, four group bars, seven parts on tap, band with likely dot and "Sharpen" while material fields are unknown, the typical household line, no letter), `EpcChip` ("Estimated EPC", spread either side until `bandValidated`). M-B: `Futures` (fixed order, four figures plus "More", `Negatives` on every card even when empty, withheld and same-as states, and `DoNothingLine` "If nothing is done" with wear effects, decay sources, grade badges, replacements, negatives and models that do not apply), `MeasureCard` (six states with pattern and label), `FuelBars` (diverging, hatch for increases, range band, table toggle), `Slider` (kumiko bar, discrete evidence stops, keys and step buttons), `WhyNot` (rule, memory, tenure, witness, target, budget), `CheckLane` (names the check, cost with `GradeBadge`, record done), `MonthRing` (twelve wedges stacked per fuel with hatches, arrow keys wrap, tap or drag, table), plus `ChartFrame`/`TableAlt`, `GradeBadge`, `Mon` and `MONS`/`isMonId`/`monFor` (WP-07 ask), `magnitude`. All props carry `EvidenceId`s; every number is a `Num`. Adapters: `negativeItems(core Negative[], text)` maps WP-02 negatives. Stories under `Kit/*` (`stories/kit-evidence.ts` is illustrative only). Not built: the rest of DS 8 (M-D), a Latin-first fonts entry (WP-09 ask, next), axe runs over stories.

**builder:WP-02** handoff · done

**M-B pieces without new physics. What exists:** `packages/engine/src/finance/{returns,owner}.ts` (simple and discounted paybacks, IRR only on one sign change by integer bisection, exact loan annuity and cash-positive flag, escalation scenarios, A_cf); `src/score/` (exact rational score, sub-scores and four groups from spec-gen constants, inputs from the standard run, estimated EPC band with one band either side); `src/flips/` (every crossing, B9 two-flip slab fixture passes); `src/search/{extras,do-nothing}.ts` and `src/evidence/outcomes.ts` (outcome nodes, per-package negatives, the "If nothing is done" run); `api.ts` implements `flips`, `doNothing`,

`negatives`; the band target works. `packs/money-uk`: costs for every SPEC 9 option and the like-for-like boiler, loan stub, `epc-proxy/rating`. **Tests:** `packages/engine/test/mb.test.ts` (B4 weight and IRR injections, B5 dominance property, B9, AC-B1, AC-B3), engine 136 and money-uk 16 pass on committed physics. **Stubbed:** T needs 5R1C comfort (score is a range until then), renewable fraction from export only, reference prices in the engine, uncertainty, SEG, value range nodes, disruption negatives, real-scale fixture (needs WP-01 measures). See the WP-02 interface entries.

**builder:WP-03** handoff · done

**Rules, gate M-B (C1 to C7).** What exists: `packages/engine/src/rules/` adds `fns/` (`mcs020a_Lp_le` rational enclosure mirroring Lean `Noise`, `adf_category` mirroring Lean `Adf`, `mees_cap`, `dt_hosting`, `ens_retv`, `noise()` fast path from site geometry), `derive.ts`, full RP 4 `homeFacts`, `obligations.ts`, `render.ts`, `lint.ts`, built-ins (budget, tenure, prerequisite, conflicts). `packs/rules-uk` has 50 rules: `ENG.PD.*` (heat pump noise, volume, count, highway, listed, roof, cooling, retired 1 m boundary, Article 4, PV, EWI), `ENG.LB.ANY`, `ENG.BR.*` (renovation with 0.70 as trigger, flow 55, sizing, AD F categories, overheating advisory, traditional walls), `ENG.MOIST.IWI` (+ costed assessment), `UK.DNO.{HP,HP.LOOPED,G98,G99}`, `UK.PAS2035`, `UK.BUS`, `EW.MEES.{NOW,CAP}`, `APPL.*` (5), Wales and Scotland stubs; every rule graded and sourced, reviewer-memory rules cite DEC SC-3 or SC-4. Tests: engine rules 45 (C1 golden enclosure inside 31.95 to 32.05 dB, upper end open, C3, C4 adf monotone and obligations idempotent, C6 noise p95 under 5 ms, renderer, lint), rules-uk 72 (one golden per rule and built-in, the six SPEC 9 acceptance cases, C5 retirement on 2025-05-29, C7 lint and sentences, GH1 M-0). Stubbed: Wales and Scotland rule bodies, `rules-in` (T3). Needs others: see the WP-03 M-B interface entry (WP-02 `noise` wiring, obligations, fixture regeneration; WP-01 option facts; WP-04 certificate env).